## Configure Autograder

Every Autograder using Gradescope must include the following Files:

1. *requirements.txt*
2. *run_autograder*
3. *run_test.py*
4. *setup.sh*
5. *tests/test_subprocess.py*

Each Test Case has Five Main Parts:

1. **Weight:** This is how many points the test case is worth and is defined using a precondition before the function
2. **Name:** There are two names for every test case. A Python unit test name and Gradescope test name (Shown to the students). The Python function name must start with test_ to indictate that it is to be tested. Every Test Case must have a different name.
3. **Subprocess:** This is the multithreaded component. Here we use Popen to create a new thread running our students code. Denote the parameter requires a list of strings therefore any command line input must be placed within their own string.
4. **Python Unit Test:** Compare the students output to an expected output using python operators.
5. **Error Message:** (Optional but Recommended): This allows you to give your students more descriptive feedback based on their output or give them common troubleshooting techniques

Submit zip to Gradescope to configure for labs.

## Student Submission

- Students can upload either **Zip** or **individual files**
- Make file is mandatory for the student to submit if instructor have not provided in *run_autograder*

## Testing:

Given: Hello World
Expected: Hello World!




Given: Hello  World!

Expected: Hello World!




Given: Hello World!

Expected: Hello World!




Test case with different file name:

### Autograder Output (hidden from students)

```
cp: missing destination file operand after '/autograder/source/lab1.cpp'
Try 'cp --help' for more information.
make: *** No rule to make target 'lab1.cpp', needed by 'default'.  Stop.
```

### Clean Compile (0/0)

```
Test Failed: False is not true : make: *** No rule to make target 'lab1.cpp', needed by 'default'.  Stop.
```

### Run: Character Count (0/5)

```
Test Failed: [Errno 2] No such file or directory: './test1.out'
```

### Run: Output Matching (0/5)

```
Test Failed: [Errno 2] No such file or directory: './test1.out'
```

17 Days, 17 Hours Late

**Student**
Unknown Student (removed from roster?)

**Total Points**
**- / 11 pts**

**Autograder Score**
**0.0 / 10.0**

**Failed Tests**
Clean Compile (0/0)
Run: Character Count (0/5)
Run: Output Matching (0/5)

**Question 2**
Auto2 1 pt